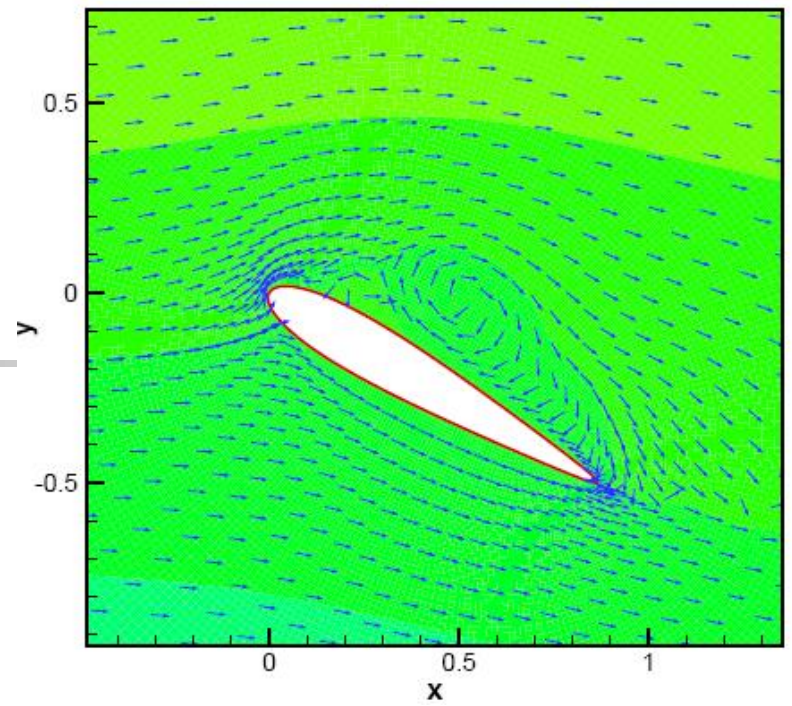
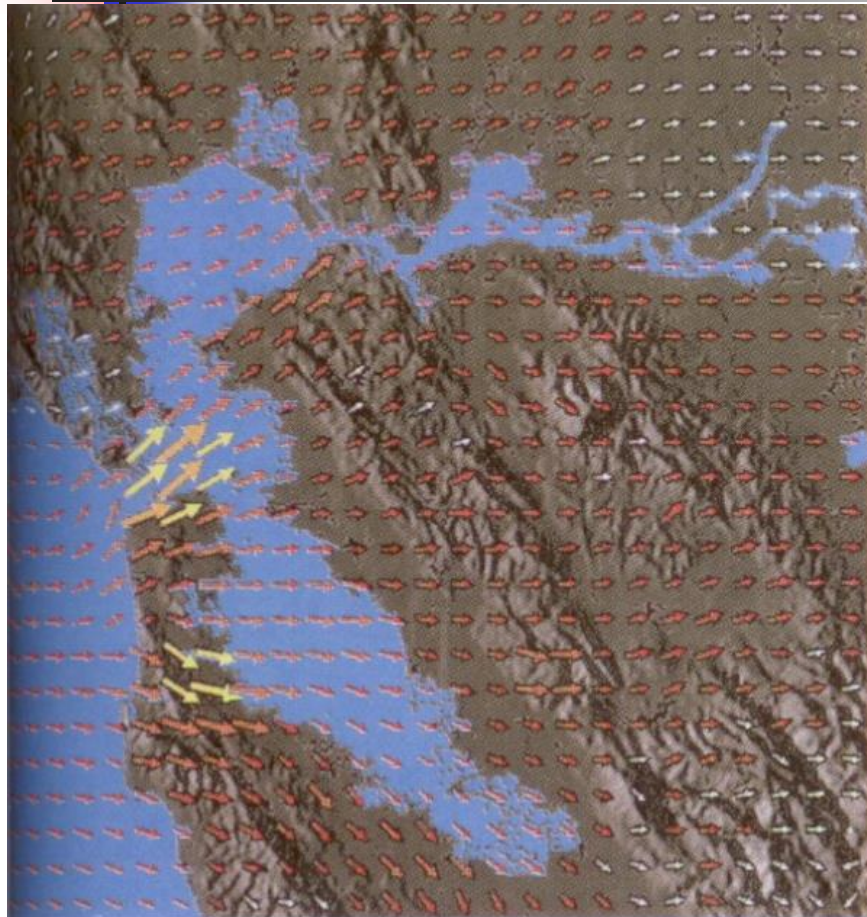


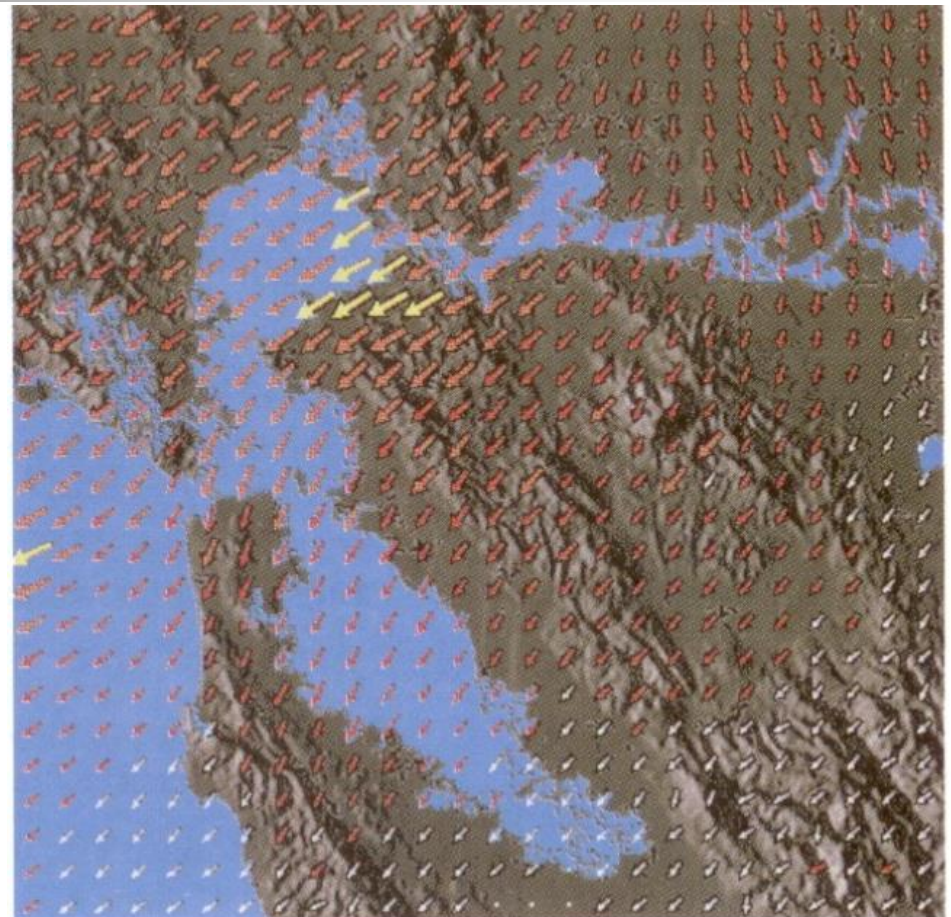


31 October 2023

VECTOR FIELDS



(a) 5:00 P.M., August 6, 1997



(b) 1:00 P.M., January 7, 1998

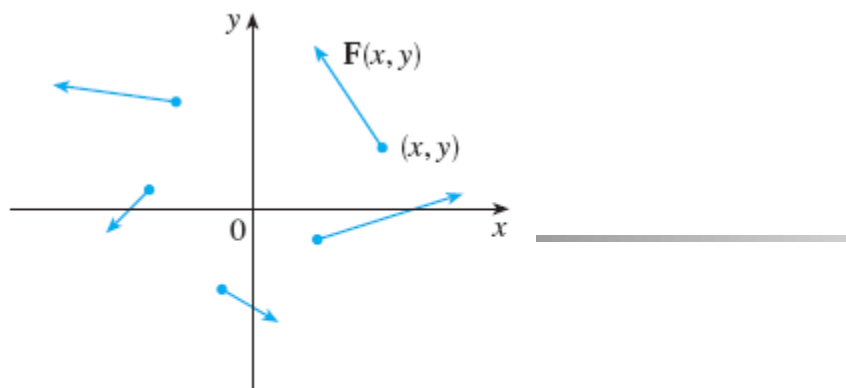
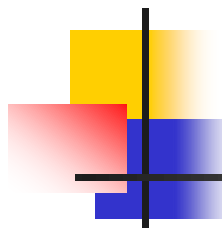


FIGURE 3
Vector field on $\mathbb{R}^2$

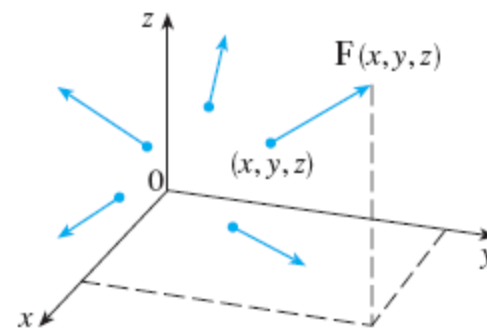


FIGURE 4
Vector field on $\mathbb{R}^3$

2 Definition Let E be a subset of $\mathbb{R}^3$. A **vector field on $\mathbb{R}^3$** is a function F that assigns to each point (x, y, z) in E a three-dimensional vector $F(x, y, z)$.

V EXAMPLE 1 A vector field on $\mathbb{R}^2$ is defined by $F(x, y) = -y\mathbf{i} + x\mathbf{j}$. Describe F by sketching some of the vectors $F(x, y)$ as in Figure 3.

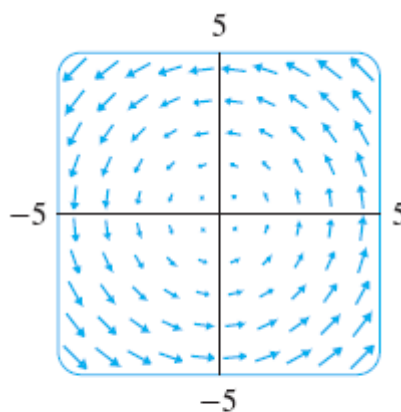
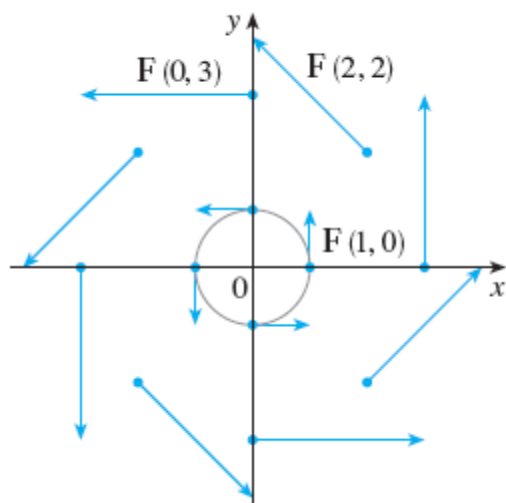


FIGURE 6
 $F(x, y) = \langle -y, x \rangle$

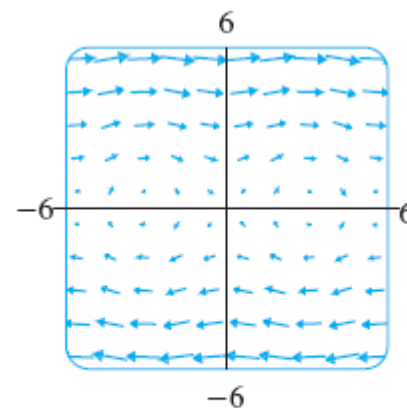


FIGURE 7
 $F(x, y) = \langle y, \sin x \rangle$

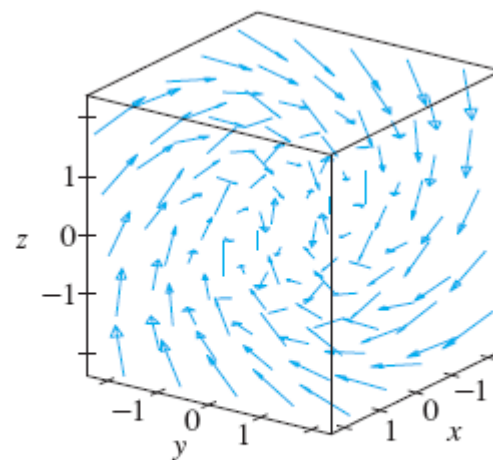
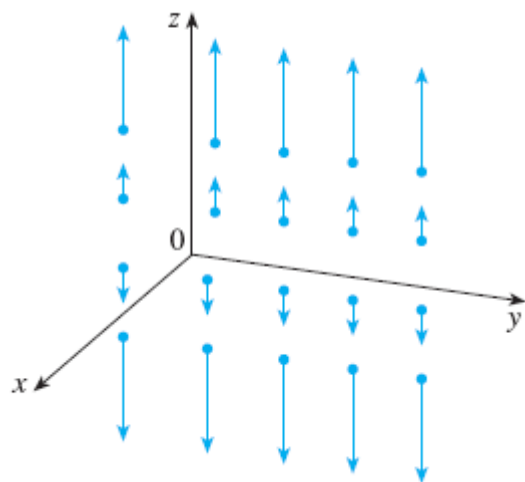


FIGURE 10
 $F(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$

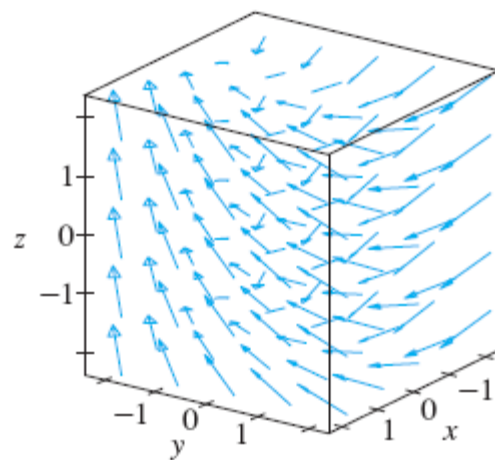


FIGURE 11
 $F(x, y, z) = y\mathbf{i} - 2z\mathbf{j} + x\mathbf{k}$

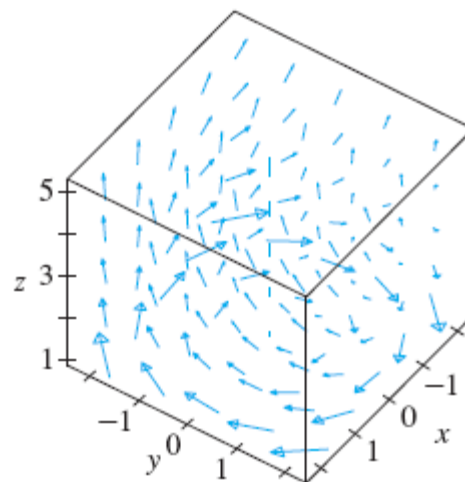
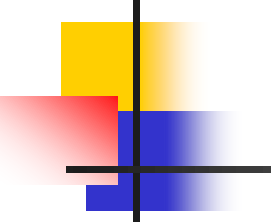


FIGURE 12
 $F(x, y, z) = \frac{y}{z}\mathbf{i} - \frac{x}{z}\mathbf{j} + \frac{z}{4}\mathbf{k}$



Gradient Fields

If f is a scalar function of two variables, recall from Section 14.6 that its gradient ∇f (or $\text{grad } f$) is defined by

$$\nabla f(x, y) = f_x(x, y) \mathbf{i} + f_y(x, y) \mathbf{j}$$

Therefore ∇f is really a vector field on $\mathbb{R}^2$ and is called a **gradient vector field**. Likewise, if f is a scalar function of three variables, its gradient is a vector field on $\mathbb{R}^3$ given by

$$\nabla f(x, y, z) = f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k}$$

Line Integrals

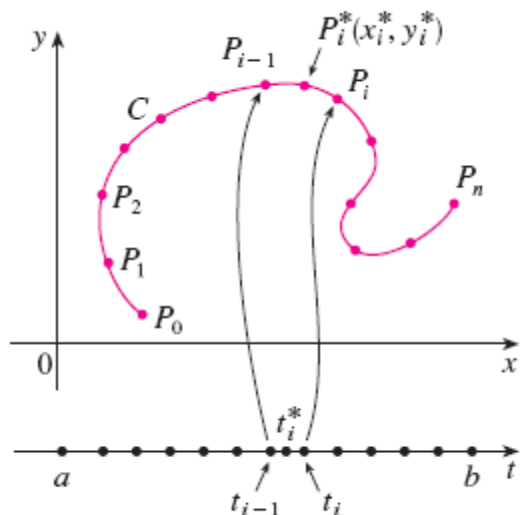
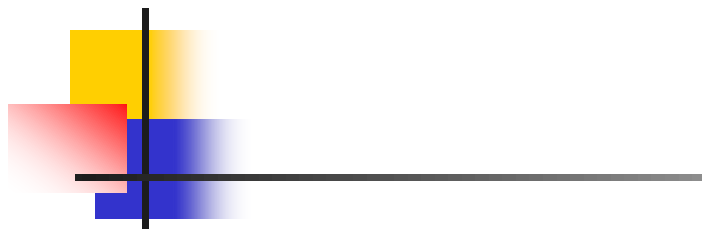
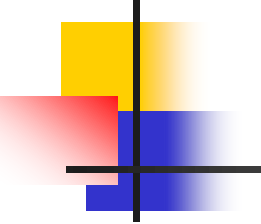


FIGURE 1

We define an integral that is similar to a single integral except that instead of integrating over an interval $[a, b]$, we integrate over a curve C . Such integrals are called *line integrals*, although “curve integrals” would be better terminology.

We start with a plane curve C given by the parametric equations

$$\boxed{1} \quad x = x(t) \quad y = y(t) \quad a \leq t \leq b$$



or, equivalently, by the vector equation $\mathbf{r}(t) = x(t) \mathbf{i} + y(t) \mathbf{j}$, and we assume that C is a smooth curve. [This means that $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$. See Section 13.3.] If we divide the parameter interval $[a, b]$ into n subintervals $[t_{i-1}, t_i]$ of equal width and we let $x_i = x(t_i)$ and $y_i = y(t_i)$, then the corresponding points $P_i(x_i, y_i)$ divide C into n subarcs with lengths $\Delta s_1, \Delta s_2, \dots, \Delta s_n$. (See Figure 1.) We choose any point $P_i^*(x_i^*, y_i^*)$ in the i th subarc. (This corresponds to a point t_i^* in $[t_{i-1}, t_i]$.) Now if f is any function of two variables whose domain includes the curve C , we evaluate f at the point (x_i^*, y_i^*) , multiply by the length Δs_i of the subarc, and form the sum

$$\sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$$

which is similar to a Riemann sum. Then we take the limit of these sums and make the following definition by analogy with a single integral.

2 Definition If f is defined on a smooth curve C given by Equations 1, then the **line integral of f along C** is

$$\int_C f(x, y) \, ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$$

if this limit exists.

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

3

$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

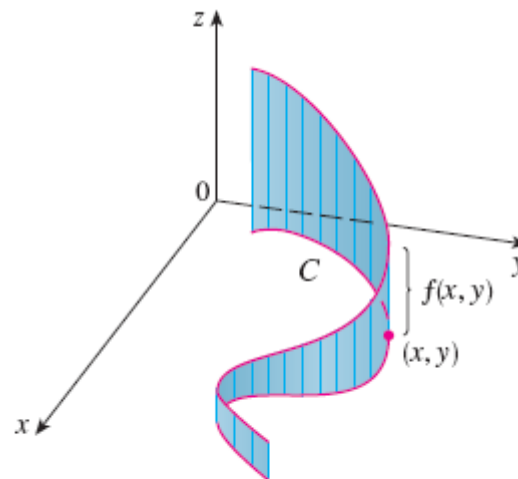


FIGURE 2

An Example

EXAMPLE 1 Evaluate $\int_C (2 + x^2 y) \, ds$, where C is the upper half of the unit circle $x^2 + y^2 = 1$.

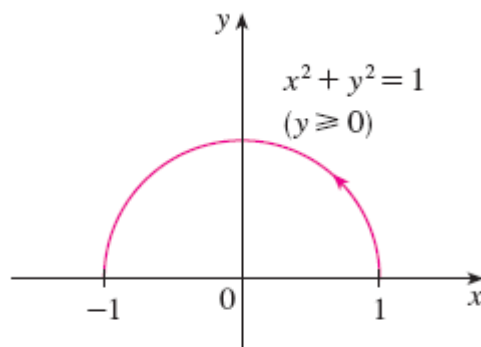


FIGURE 3

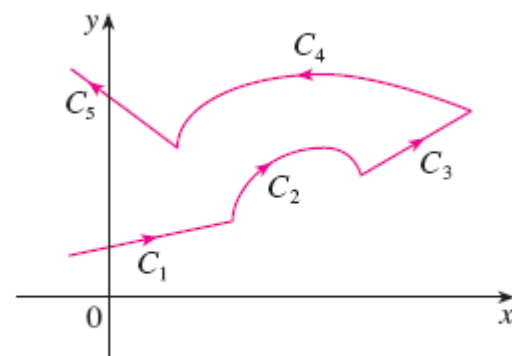
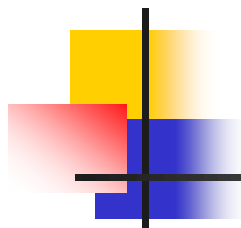


FIGURE 4
A piecewise-smooth curve

Suppose now that C is a **piecewise-smooth curve**; that is, C is a union of a finite number of smooth curves $C_1, C_2, \dots, C_n$, where, as illustrated in Figure 4, the initial point of C_{i+1} is the terminal point of C_i . Then we define the integral of f along C as the sum of the integrals of f along each of the smooth pieces of C :

$$\int_C f(x, y) \, ds = \int_{C_1} f(x, y) \, ds + \int_{C_2} f(x, y) \, ds + \cdots + \int_{C_n} f(x, y) \, ds$$

An Example

EXAMPLE 2 Evaluate $\int_C 2x \, ds$, where C consists of the arc C_1 of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$ followed by the vertical line segment C_2 from $(1, 1)$ to $(1, 2)$.

$$= \frac{5\sqrt{5} - 1}{6} + 2$$

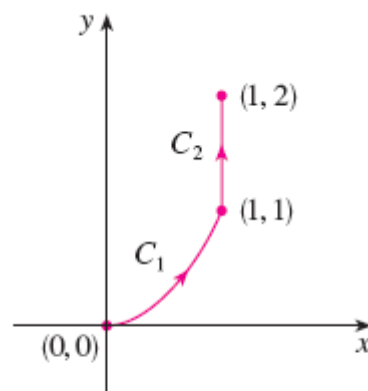
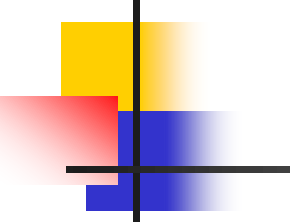


FIGURE 5
 $C = C_1 \cup C_2$

Mass & Center of Mass of Wire



Any physical interpretation of a line integral $\int_C f(x, y) \, ds$ depends on the physical interpretation of the function f . Suppose that $\rho(x, y)$ represents the linear density at a point (x, y) of a thin wire shaped like a curve C . Then the mass of the part of the wire from P_{i-1} to P_i in Figure 1 is approximately $\rho(x_i^*, y_i^*) \Delta s_i$ and so the total mass of the wire is approximately $\sum \rho(x_i^*, y_i^*) \Delta s_i$. By taking more and more points on the curve, we obtain the **mass** m of the wire as the limiting value of these approximations:

$$m = \lim_{n \rightarrow \infty} \sum_{i=1}^n \rho(x_i^*, y_i^*) \Delta s_i = \int_C \rho(x, y) \, ds$$

[For example, if $f(x, y) = 2 + x^2 y$ represents the density of a semicircular wire, then the integral in Example 1 would represent the mass of the wire.] The **center of mass** of the wire with density function ρ is located at the point $(\bar{x}, \bar{y})$, where

$$\boxed{4} \quad \bar{x} = \frac{1}{m} \int_C x \rho(x, y) \, ds \quad \bar{y} = \frac{1}{m} \int_C y \rho(x, y) \, ds$$

Other physical interpretations of line integrals will be discussed later in this chapter.

V EXAMPLE 3 A wire takes the shape of the semicircle $x^2 + y^2 = 1, y \geq 0$, and is thicker near its base than near the top. Find the center of mass of the wire if the linear density at any point is proportional to its distance from the line $y = 1$.

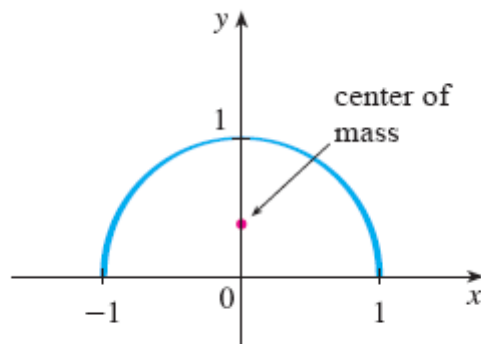
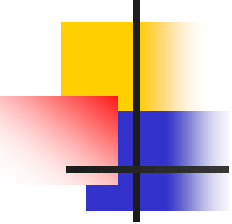


FIGURE 6

Line integrals w.r.t. x and y



Two other line integrals are obtained by replacing Δs_i by either $\Delta x_i = x_i - x_{i-1}$ or $\Delta y_i = y_i - y_{i-1}$ in Definition 2. They are called the **line integrals of f along C with respect to x and y** .

$$\boxed{5} \quad \int_C f(x, y) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta x_i$$

$$\boxed{6} \quad \int_C f(x, y) dy = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta y_i$$

When we want to distinguish the original line integral $\int_C f(x, y) ds$ from those in Equations 5 and 6, we call it the **line integral with respect to arc length**.

The following formulas say that line integrals with respect to x and y can also be evaluated by expressing everything in terms of t : $x = x(t)$, $y = y(t)$, $dx = x'(t) dt$, $dy = y'(t) dt$.

$$\boxed{7} \quad \begin{aligned} \int_C f(x, y) dx &= \int_a^b f(x(t), y(t)) x'(t) dt \\ \int_C f(x, y) dy &= \int_a^b f(x(t), y(t)) y'(t) dt \end{aligned}$$

V EXAMPLE 4 Evaluate $\int_C y^2 dx + x dy$, where (a) $C = C_1$ is the line segment from $(-5, -3)$ to $(0, 2)$ and (b) $C = C_2$ is the arc of the parabola $x = 4 - y^2$ from $(-5, -3)$ to $(0, 2)$. (See Figure 7.)

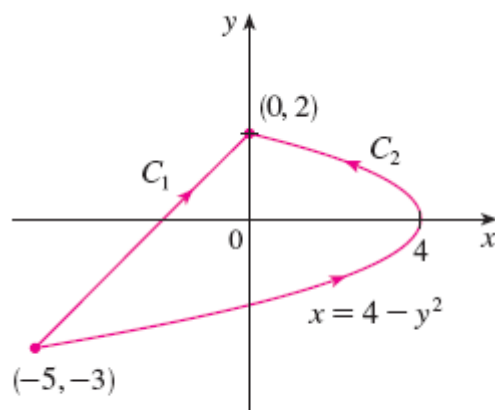


FIGURE 7

Orientation

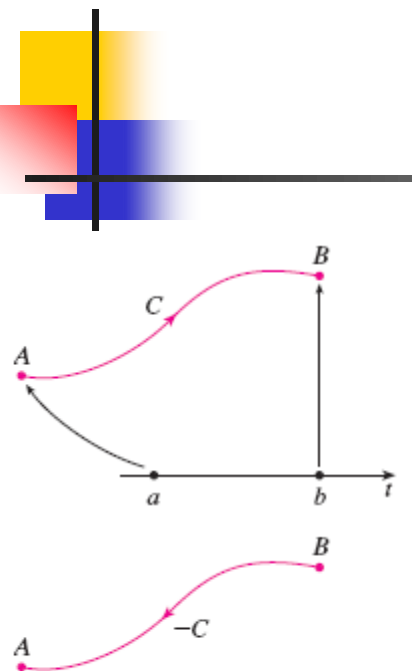


FIGURE 8

In general, a given parametrization $x = x(t)$, $y = y(t)$, $a \leq t \leq b$, determines an **orientation** of a curve C , with the positive direction corresponding to increasing values of the parameter t . (See Figure 8, where the initial point A corresponds to the parameter value a and the terminal point B corresponds to $t = b$.)

If $-C$ denotes the curve consisting of the same points as C but with the opposite orientation (from initial point B to terminal point A in Figure 8), then we have

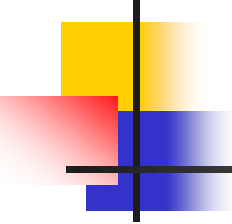
$$\int_{-C} f(x, y) dx = -\int_C f(x, y) dx \quad \int_{-C} f(x, y) dy = -\int_C f(x, y) dy$$

But if we integrate with respect to arc length, the value of the line integral does *not* change when we reverse the orientation of the curve:

$$\int_{-C} f(x, y) ds = \int_C f(x, y) ds$$

This is because Δs_i is always positive, whereas Δx_i and Δy_i change sign when we reverse the orientation of C .

Line Integrals in Space



We now suppose that C is a smooth space curve given by the parametric equations

$$x = x(t) \quad y = y(t) \quad z = z(t) \quad a \leq t \leq b$$

or by a vector equation $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$. If f is a function of three variables that is continuous on some region containing C , then we define the **line integral of f along C** (with respect to arc length) in a manner similar to that for plane curves:

$$\int_C f(x, y, z) \, ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*, z_i^*) \Delta s_i$$

We evaluate it using a formula similar to Formula 3:

$$\boxed{9} \quad \int_C f(x, y, z) \, ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt$$

An Example

V EXAMPLE 5 Evaluate $\int_C y \sin z \, ds$, where C is the circular helix given by the equations $x = \cos t$, $y = \sin t$, $z = t$, $0 \leq t \leq 2\pi$. (See Figure 9.)

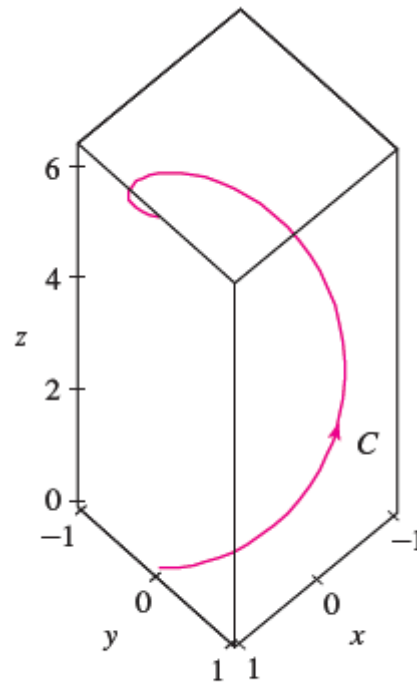


FIGURE 9